DATA ENGINEERING COURSE

Advanced Functions

ASSIGNMENT

*args **kwargs Lambda map() filter()

Instructor: **Tanishq Tyagi**

Attempt all questions independently and test your code thoroughly

Student Instructions

- Read all questions carefully before you begin writing code.
- Attempt all questions independently — do not copy or share solutions.
- Test your code thoroughly with multiple inputs, including edge cases.
- Follow Python coding best practices: meaningful variable names, clean indentation, comments where needed.
- Do not use any external libraries — all questions are solvable with core Python.
- Submit your final .py file(s) as instructed by your instructor.

A *args — Arbitrary Positional Arguments

Q1 Write a function that accepts any number of integers as arguments and returns their sum.

Q2 Write a function that accepts any number of numbers as arguments and returns the maximum number among them.

Q3 Write a function that accepts any number of student names as arguments and displays (prints) each name.

B **kwargs — Arbitrary Keyword Arguments

Q4 Write a function that accepts employee details as keyword arguments and prints each detail in the format: key: value

C Lambda Functions

Q5 Write a lambda function that accepts a number and returns its square.

Q6 Write a lambda function that accepts three numbers and returns their product (multiplication).

Q7 Write a lambda function that accepts a number and returns True if the number is even, or False if it is odd.

D map() — Applying Transformations

Q 8	Given a list of numbers, use map() to return a new list where every number is doubled.
Q 9	Given a list of names (which may be in mixed case), use map() to convert every name to uppercase and return the result as a list.

E filter() — Selective Filtering

Q 10	Given a list of numbers, use filter() to return a new list containing only the even numbers.
Q 11	Given the following list of salaries, use filter() to return only the salaries that are greater than 50,000: salaries = [30000, 55000, 45000, 72000, 68000, 25000, 90000, 48000]

All the best! Refer to your class notes if needed, but attempt each question on your own first.